

Where Optimal Networks Stop Existing

Certified Spectral Limits of Cyclic Architectures

Category: Mathematics

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Every graph has a zeta function, so every graph has a Riemann Hypothesis — and unlike the classical one, it can be **decided**. We decide it **completely and in exact arithmetic** for the generalized Petersen graphs, and prove that only **finitely many** satisfy it.

INTRODUCTION — an RH you can decide

Every graph has a zeta function. The Ihara zeta

$\zeta_X(u) = \prod_{[C]} (1 - u^{\ell(C)})^{-1}$, over primitive closed geodesics, has an Euler product, a functional equation, and a **prime number theorem**.

So one can ask the RH for a graph: do the poles of $\zeta_X(q^{-s})$ in the critical strip lie on $\operatorname{Re} s = \frac{1}{2}$?

And here it is decidable. By the Ihara determinant formula, a $(q+1)$ -regular graph satisfies the RH *iff* it is **Ramanujan**: every eigenvalue with $|\lambda| \neq q+1$ has $|\lambda| \leq 2\sqrt{q}$.

$2\sqrt{q}$ is the smallest gap any infinite family can have: the **optimal expanders**.

INTRODUCTION — the question

For which n and k does $P(n, k)$ satisfy the Riemann Hypothesis?

A question of a recognised kind. The same classification exists for unitary Cayley graphs (Droll) and integral circulants (Le-Sander) — only *finitely many* members qualify.

The ingredients have been on the shelf since 2011. Gera and Stănică determined the spectrum of $P(n, k)$ completely — but their paper contains **no occurrence** of *Ramanujan*, *expander* or *spectral gap*.

And it is not just a computation. At a band edge the decision compares nested radicals, where floating point does not decide — it guesses.

FRAMEWORK — a symbol on a grid

$P(n, k)$ is a **cyclic cover**: it carries the rotation $\rho : i \mapsto i + 1$, so the adjacency matrix decomposes over the characters of $\mathbb{Z}_n$:

$$M_j = \begin{pmatrix} \alpha_j & 1 \\ 1 & \beta_j \end{pmatrix}, \quad \alpha_j = 2 \cos \frac{2\pi j}{n}, \quad \beta_j = 2T_k(\alpha_j/2).$$

On the zeta side this is the factorization into Artin–Ihara L -functions.

So the spectrum is **one fixed curve sampled at n points**, and the RH holds when no sample lands where the curve is too big. **Does the grid miss the bad bands?** It refines as n grows — which will bound n .

FINDINGS 1 — clearing the radicals

Eigenvalues $(A \pm \sqrt{(\alpha - \beta)^2 + 4})/2$ against $2\sqrt{2}$: **two nested irrationalities**. Substituting $\lambda = \pm 2\sqrt{2}$ into $\lambda^2 - A\lambda + (P - 1)$ gives $2\sqrt{2}|A| \leq 7 + P$. Squaring twice removes both radicals, and **each squaring is an equivalence**: $|A| \leq 4 < 4\sqrt{2}$, $|P| \leq 4$.

Criterion. $P(n, k)$ satisfies the RH iff

$$D_k(u) = (7 + 4uT_k(u))^2 - 8(2u + 2T_k(u))^2 \geq 0$$

at $u = \cos \frac{2\pi j}{n}$ for every non-trivial j .

The poles of a zeta function: now **one integer polynomial**.

FINDINGS 2 — finiteness

Everything follows from three **exact** evaluations:

$$D_k(1) = -7, \quad D_k(0) \geq 17, \quad D_k(-1) = -7 \text{ (} k \text{ odd), } +9 \text{ (} k \text{ even).}$$

$D_k(1) < 0$ puts a **forbidden band at** $u = 1$, where the grid accumulates. With $s = (1 - \cos t) + (1 - \cos kt)$, AM–GM gives $4ab \leq s^2$, and $s^2 + (4\sqrt{2} - 4)s$ hits $8\sqrt{2} - 11$ **exactly** at $s = (\sqrt{2} - 1)^2$; then $1 - \cos \theta \leq \theta^2/2$ bounds t .

Theorem — for every k . $n \leq \pi(2 + \sqrt{2})\sqrt{k^2 + 1}$. No expansion, no computation. Per k the exact root sharpens this to $B_k = 2\pi / \arccos u_k$, **sharp at** $k = 2, 3, 4$.

FINDINGS 3 — a parity law

For *odd* k , $D_k(-1) = -7 < 0$: a **second band at $t = \pi$** . For even k , $D_k(-1) = +9 > 0$ and there is none.

But the grid meets $t = \pi$ only when n is even — and there the sample is the -3 of a *bipartite* graph, which the Ramanujan condition **exempts**. Odd n lands just short, inside the band.

Theorem (parity law). At $t = \pi$ the same algebra returns the *identical* expression, so odd n is cut at **exactly half** the same constant. **Sharp at $k = 1, 5, 7, 9$.**

Being bipartite helps — worth a factor of two in n .

FINDINGS 4 — a theorem for $k \leq 9$, exact arithmetic beyond

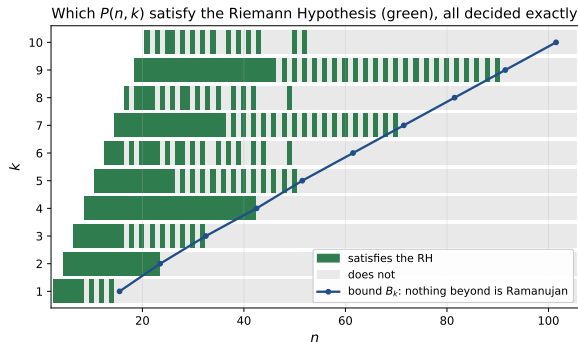
Theorem (closed form). For $k \in \{1, 2, 3, 4, 5, 7, 9\}$ the negative set of D_k on $[-1, 1]$ is the end band(s) alone — Sturm's theorem on one integer polynomial — so only the grid point $j = 1$ (and, for odd n and odd k , $j = \frac{n-1}{2}$) can fail:

$$P(n, k) \text{ Ramanujan} \iff 2k < n \leq B_k, \quad \text{and } n \leq B_k/2 \text{ if } n, k \text{ odd.}$$

From $k = 10$ on an interior band always appears, and the list has gaps.

Beyond $k = 9$: D is an algebraic integer of $\mathbb{Z}[\zeta_m]$, decided exactly mod Φ_m — a finite check after a theorem made the problem finite.

FINDINGS 5 — the complete classification



Theorem. Every n , every k : exactly 460 pairs satisfy the RH (324 up to isomorphism), from 13110 cases in **exact arithmetic**. Nothing past $k = 45$ qualifies.

FINDINGS 6 — the same engine

The same decomposition settles the **inverse eigenvalue problem** here: over *every* matrix carrying the pattern, eliminating the inner coordinates leaves one recurrence of order $2k + 2$, so $\text{null } A \leq 2k + 2$.

That ceiling equals the best known upper bound on Z — so the method either settles it completely or cannot settle it at all.

Building the matrix from interchangeable **tiles** turns an all- n claim into finitely many finite ones. Certifying 77 tiles exactly gives $Z(P(n, k)) = M(P(n, k)) = 2k + 2$ for **every** n at $k = 2, 3, 4$ — proving **Conjecture 5** of a 2026 note.

CONCLUSIONS

How this answers the question. Clearing the radicals gives one integer polynomial; it factors into a form with *no k in it*; Dirichlet bounds n uniformly in k ; for $k \leq 9$ ($k \neq 6, 8$) the answer is a theorem, and the finite remainder is decided in exact arithmetic.

Proved: criterion, corner form, finiteness, parity, closed form, ceiling, symbol classification, tiling, the leading-coefficient criterion. **Finite exact checks:** the census for $k \geq 10$, small Z . **One computer-assisted existence proof:** the tiles.

Open. $M(P(24, 4))$; an effective $L(k)$ and a tile at $k \geq 6$; the cover ceiling when $c_{\max}$ has two monomials; a hand proof of tile existence.

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